

- 5** Analysis of a mixture of sulfur-containing gases shows that hydrogen sulfide, H_2S , and carbon disulfide, CS_2 , are present in a 2 : 1 mole ratio.

60 cm^3 of the gaseous mixture is burned in excess oxygen. The resulting gases are bubbled through aqueous sodium hydroxide. All volumes are measured under the same temperature and pressure.

Which row correctly describes

- the SO_2 : CO_2 mole ratio in the gaseous mixture obtained after complete combustion and
- the reduction in volume of the gaseous mixture when bubbled through $\text{NaOH}(\text{aq})$?

	SO_2 : CO_2 mole ratio	reduction in volume
A	3 : 1	80 cm^3
B	4 : 1	80 cm^3
C	3 : 1	100 cm^3
D	4 : 1	100 cm^3